

Program : Diploma in Polymer Technology	
Course Code : 6092C	Course Title: Paints and Surface Coatings
Semester : 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- Understand the properties of light
- Identify the components of paints, its preparations and application
- Understand the different methods of Painting and adhesion of coatings
- Know the mechanism of film formation and the various properties of coatings

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (hours)	Cognitive level
CO1	Explain the properties of light	14	Understanding
CO2	Identify different types of paints, components of paints and its preparations	14	Understanding
CO3	Apply the different methods of Painting and adhesion of coating	16	Applying
CO4	Know the mechanism of film formation and the various properties of coatings	14	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		2				
CO3	3		2				
CO4	3		2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Explain the properties of light		
M1.01	Summarize the properties of light	3	Remembering
M1.02	Identify the components of paint	3	Understanding
M1.03	Describe types of solvents and solvent properties	4	Understanding
M1.04	Explain the preparation of paint	4	Understanding
Contents: Light: Reflection, Refraction, Diffraction, Color Science, Additive Color Mixing, Subtractive Color Mixing, Gloss - Introduction: Components of Paints - Pigments - Manufacture - Pigment Properties - Different Types- Selection, Dispersion and Color Matching of Pigments - Pigment Dispersion and Preparation, Extenders, Solvents, Different Types, Solvent Properties, Oil, Driers, Resins, Diluents, Additives Affecting: Viscosity Interfacial Tensions - Paint Preparation - Formulation			
CO2	Identify the different types of paints, components of paints, preparations and applications		
M2.01	List different types of paints	3	Remembering
M2.02	Distinguish between oil paint and emulsion paint	3	Understanding
M2.03	Explain the classification of paints based on resins used.	4	Understanding
M2.04	Explain the classification of paints based on application.	4	Understanding
	Series Test – I	1	

Contents: Classification Based on Polymeric Resin, Alkyd Paints, Acrylic , Epoxy Coatings, Polyurethanes, Silicones, Formaldehyde based Resins, Classification based on Application, Appliance Finishes, Automotive Finishes, Marine Coatings, and Aircraft Finishes.			
CO3	Apply the different methods of Painting and adhesion of coating		
M3.01	Explain the methods adopted for surface cleaning and its effect on adhesion Properties of Coatings	4	Understanding
M3.02	Describe various methods adopted for application of paints	4	Understanding
M3.03	Explain adhesion Properties of Coatings	4	Understanding
M3.04	Explain the factors Affecting Adhesive Bond	4	Understanding
Contents: Surface Cleaning Methods, Chemical Conversion Treatments, Paint Application, Brushing, Dip Coating, Flow Coating, Roller Coating, Spray Painting, Electro deposition, Chemiphoretic Deposition- Adhesion Properties of Coatings, Properties such as Floating, Silking, Cratering, -Factors Affecting viscosity, Adhesion bond			
CO4	Know the mechanism of film formation and the various properties of coatings		
M4.01	Explain the mechanism of film formation	3	Understanding
M4.02	Explain the factors affecting coating properties.	3	Understanding
M4.03	Explain the properties of coatings	4	Understanding
M4.04	Explain the optical properties and ageing properties	4	Understanding
	Series Test – II	1	
Contents: Mechanism of Film Formation, Physical Drying, Oxidation Drying, Chemical Drying, Factors Affecting Coating Properties, Film Thickness, Film Density, Internal Stresses, Pigment Volume Concentration (PVC). Different Methods used for Film Preparation, Barrier Properties, Mechanical Properties and Optical Properties of Coatings, Color, Gloss, Hiding Power, Ageing Properties, Factors Affecting Viscosity of Paints.			

Text / Reference:

T/R	Book Title/Author
T ₁	Swaraj Paul, “Surface Coatings: Science and Technology”, Wiley – Interscience 1985.
R ₁	R.Lambourne. “Paint and Surface Coatings – Theory and Practice”, Ellis Horwood Chichester 1987
R ₂	Turner, Paints and Surface Coating

Online Resource:

Sl.No	Website Link
1	https://www.elsevier.com
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